

Fixed-Wing Survey Drone

Long-Endurance Mapping & GIS Intelligence Platform

OVERVIEW

The HFSW-02 is an ultra-efficient fixed-wing UAV purpose-built for extended mapping missions across large coverage areas. Delivers centimeter-level GSD accuracy for land surveying, agriculture, and GIS data collection — maximising every battery cycle with minimal energy consumption.

TECHNICAL SPECIFICATIONS

Flight Time	120 – 180 minutes
Operational Range	100+ km
Payload	High-Resolution RGB / Multispectral Camera
Navigation	Autonomous Mission Planning
Ground Sampling Distance	Sub-3 cm GSD at 100 m AGL
Data Output	Orthomosaic, DSM, 3D Point Cloud
Airframe	Carbon Fiber + EPO Composite
Launch Method	Hand / Bungee Launch
Landing	Belly Landing / Deep-Stall
Compatibility	Mission Planner, QGroundControl, Pix4D, Agisoft

KEY FEATURES

- ▶ Mapping-grade accuracy with RTK/PPK GPS for centimeter precision
- ▶ Long endurance enables large-area coverage in a single sortie
- ▶ Automated flight path generation with obstacle avoidance
- ▶ Real-time data downlink and live map preview during flight
- ▶ Native compatibility with industry-standard photogrammetry software

USE CASES

- ▶ Land Surveying & Cadastral Mapping
- ▶ Agriculture Field Mapping
- ▶ GIS Data Collection & Analysis
- ▶ Forest & Vegetation Assessment
- ▶ Construction Site Progress Monitoring
- ▶ Infrastructure & Pipeline Inspection